F005 | External Wave Benchmark

Preflight review and local data-pilot handoff • 19 October 2026

Scope: review of user-uploaded SUMMARY.md and preflight_report.json. The underlying 129 MB and 49.9 GB datasets were not attached to this session. Thus no experimental sloshing metrics have yet been calculated here.

1. Input archives: inspected metadata

Item	Robot ZIP	Roll-tank ZIP
Size	129.2 MB	49.91 GB
Archive directory	OK	OK
Members	5	346
Files	5	299
Unsafe members	0	0
MD5 verification	NOT RUN	NOT RUN

2. Verified preflight facts

• Robot archive lists 5 files: two NPY arrays, a metadata CSV, a time-vector CSV, and the original loader script. All five are recorded as extracted in the user preflight report.

• The 49.9-GB archive contains 299 files and 47 additional directory ZIP entries. There are candidate NoBody and FloatingBody records; the ZIP is still source-only.

• Both source MD5 values remain UNVERIFIED; a working ZIP central directory does not by itself establish download integrity.

3. Pilot action

Run in the user F005-v0.1.0 directory:

```
py -3 .\data\f005_data_pilot_v0.2.0.py --root . --inspect-large --verify-small-md5
```

The adapter maps robot pairs in-place, checks units and time data, creates descriptive traces and reads just one small NoBody pressure CSV from the large ZIP. It does not unzip 49.9 GB.

Current physical status: UNRESOLVED | DEVELOPMENT / NOT CERTIFIED

Qualification and provenance boundaries

The data-pilot instrument separates provenance gates, measurement-schema gates, and physical falsification. No layer is promoted by another layer passing.

P0 ZIP directory and safe names | PASS (source preflight)

Directory inspection and ZIP member names only.

P1 Source integrity | UNRESOLVED

Compute source MD5 and compare with published Zenodo values.

P2 Paired robot arrays | LOCAL RUN REQUIRED

Verify rows, 9822 x 1500 dimensions, metadata alignment and finite values.

P3 Time and units | LOCAL RUN REQUIRED

Treat time_vector.csv as authoritative; acceleration in m/s², elevation in mm.

P4 NoBody pressure | LOCAL RUN REQUIRED

CRC check a single CSV, then independently confirm units and acquisition columns.

P5 Split and leakage | NOT RUN

Provisional partition must not be called a blind split.

P6 Conventional fluid model | NOT RUN

Out-of-sample classical sloshing baseline, with stated uncertainty.

P7 PKLSA-native prediction | NOT RUN

Independent, source-native calculation without fitting test measurements.

P8 Physical SST verdict | UNRESOLVED

No Kelvin, inertial, gravity or Swirl Clock claim from sloshing data alone.

Provenance

Source evidence: supplied preflight_report.json and SUMMARY.md. External dataset DOIs: 10.5281/zenodo.20039757 and 10.5281/zenodo.15223412. This PDF is a review of metadata, not an independent verification of dataset contents or a framework-v1.0.7 certified result.

Minimum next upload: pilot_report.json, PILOT_REPORT.md, NoBody_single_pressure_preview.txt. Do not transfer either complete ZIP.